

THE BONDI DIPOLE IN FULL NUMERICAL RELATIVITY: A SELF-ACCELERATING POSITIVE–NEGATIVE MASS BINARY WITH GRTECLYN

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Bondi observed in 1957 that bodies whose active gravitational masses have opposite signs should *self-accelerate*: the negative one chases the positive one it repels, and the pair runs off together. Seven decades of analyses treated only point particles or exact metrics; here we present the first 3+1 numerical-relativity evolutions of a “Bondi dipole,” built on what is, to our knowledge, the first stable star with negative total (ADM) mass evolved in numerical relativity. Two complex scalar fields share identical dynamics; only the phantom sector sources Einstein’s equations with a minus sign: inertial and passive masses positive, active mass negative. Released at rest, the mixed pair—and only the mixed pair—self-accelerates as Bondi predicted, at 50 to 150 times the residual motion of its single-star and same-sign controls. The drift reverses when the stars swap places, scales with the phantom’s mass to 0.6%, and approaches the Newtonian rate at wider separations. In the signed bookkeeping the accelerating pair carries near-zero total momentum, as in Forward’s analysis.

Draft — required before submission: (1) convergence tests: rerun PM and one control at $N = 192, 256$, Richardson-extrapolate the drifts, quote a convergence order and numerical error bars on every headline percentage (50–150×, 0.6%, ~10%); (2) re-run the separation series with the core-weighted centroid and the per-sector momentum-balance diagnostic (Sec. VII, next steps); (3) sponge layer for the reflected radiation bath, extending the window past $t = 60$ for the constant-gap test; (4) a robustness point at a second frequency or coupling.

I. INTRODUCTION

Bondi [1] gave negative mass its modern treatment, distinguishing inertial, passive, and active gravitational mass and noting the striking consequence of mixing signs: a negative-mass body is *attracted* by a positive one, the positive body is *repelled* by the negative one, and the pair runs off together (Fig. 1)—momentum conserved, because the negative body’s contribution is negative. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated opposite-mass pairs, later generalized in Ref. [3]; Bonnor [4] asked why mass is positive and found no answer in the theory. Forward [5] analyzed the Newtonian two-body problem exhaustively—for equal and unequal mass magnitudes alike—showing “in tedious detail” that the self-accelerating pair violates neither momentum nor energy conservation; objections to negative mass must be sought elsewhere [6, 7].

Why does this deserve a full numerical-relativity campaign? Because nothing in Einstein’s equations forces mass to be positive—positivity is imposed through energy conditions on matter—and modern physics keeps producing sources that strain those conditions: Casimir energies, quantum fields in curved spacetime, a dark-energy sector whose observational fits tolerate phantom

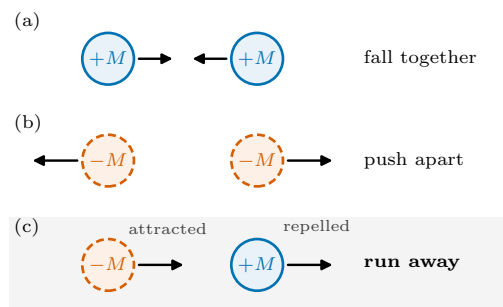


FIG. 1. Bondi’s sign rules for two bodies released at rest, in the style of Forward’s analysis [5]. Every body here has positive inertial and passive mass; only the *active* (source) mass carries a sign—negative for the dashed bodies. (a) Two positive-mass stars attract each other and fall together; (b) two negative-mass stars repel each other and push apart. In both cases the forces oppose, and the pair’s centre stays put. (c) The mixed pair—the Bondi dipole: the positive star is repelled by the negative one, the negative star is attracted by the positive one, so both forces point the *same* way and the pair runs off together, carrying zero total momentum in the signed bookkeeping. Each row is realized as a run of the matrix in Table I.

equations of state [8]. Every traversable wormhole [9] and warp spacetime [10] requires matter of exactly this kind, and the standing objection to all of them is Bondi’s runaway itself: matter that accelerates for free looks like a perpetual-motion machine, hence absurd [6, 7]. That objection has only ever been examined for point particles and special exact metrics. Whether the runaway actually happens in full nonlinear general relativity—with finite bodies, backreaction, and radiation—and whether momentum conservation really survives it, is a concrete, checkable question. Answering it also requires learning to evolve energy-condition-violating matter stably in numerical relativity at all, a capability any quantitative study of exotic spacetimes will need.

All of the analyses above, seven decades of them, treat

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point particles, prescribed metrics, or frozen mass parameters. They cannot say whether such a configuration can be posed as a *consistent initial-value problem*—constraint-satisfying initial data for dynamical matter—or whether the runaway survives full nonlinear evolution with backreaction, radiation, and finite body size. That is the gap this paper closes; we are not aware of any previous 3+1 evolution of a gravitationally bound positive-negative active-mass pair with dynamical, constraint-solved matter.

Our contributions:

- (i) A minimal field-theoretic realization of Bondi’s sign structure: a *bicomplex* scalar model in which both sectors obey identical Klein–Gordon dynamics and only the Einstein source is signed (Sec. II).
- (ii) Dressed soliton stars for both signs, solved at fixed frequency as constraint-satisfying initial data—including, as far as we know, the first star with negative total (ADM) mass evolved this way (Sec. III).
- (iii) A falsifiable run matrix—mixed pair, same-sector pairs, single stars, a mirrored pair, two wider separations—run under identical numerics and verified at $t = 0$ against the independently solved stars (Secs. IV, V).
- (iv) The result: the mixed pair, and only the mixed pair, self-accelerates—with unequal mass magnitudes (the reference pair, $|M_-|/M_+ = 1.20$) and with the magnitudes matched, echoing Forward’s unequal-mass cases; the effect scales with the source mass and converges to the Newtonian point-mass prediction as the bodies separate (Sec. V).
- (v) An honest accounting of failure modes and diagnostic biases that mimic or mask the physics under test (Secs. IIID, VI).

One more first rode along with the physics: every evolution here ran on a single GPU, with GRTeclyn [21], the AMReX-based [23] GPU port of GRChombo [20]. To our knowledge these are the first evolutions of energy-condition-violating matter in GPU-native numerical relativity, and the fact that the campaign—initial data, nine production runs, and their diagnostics—fit on one multi-GPU node is itself a statement about how accessible this class of question has become.

Throughout, $G = c = 1$ and the scalar mass sets the unit, $m = 1$. Drifts along the pair axis are quoted as $\Delta x = x(t) - x(0)$, and d denotes the instantaneous barycentre separation. The paper follows the build order: Sec. II defines the matter model, Sec. III constructs the stars and the initial data—including the failed attempts that shaped the recipe—Sec. IV lays out the runs, Sec. V the measurements, and Secs. VI–VII their limits and meaning.

II. THE BICOMPLEX SCALAR MODEL

We first need matter that can realize Bondi’s sign structure while remaining an honest field theory, and the requirements are stiffer than they look. The bodies must keep canonical *dynamics*—positive inertial and passive mass—while only their gravitational *source* changes sign; they must form stable, localized stars; and the negative-mass star must be held together by something other than gravity, because its own gravity pushes it apart. Each obvious alternative fails one of these tests. The negative-mass Schwarzschild spacetime is a naked singularity with unstable perturbations [11], and contains no matter from which to build constraint-satisfying initial data. A negative-energy fluid has no equilibrium to sit in: a fluid star is bound by the very attraction being flipped. A ghost scalar—kinetic term reversed in the Lagrangian—does source gravity negatively, but its field dynamics reverse too, so its inertial and passive masses flip along with the active one (Forward’s all-negative matter [5], not the active-only signature under test), and such fields, evolved in numerical relativity as wormhole support, either collapse or explode [12].

What passes is a *soliton*: matter self-bound by its own interactions, for which gravity is a dressing rather than the binding agent. We therefore take two independent complex scalar fields, Φ_+ (*canonical*) and Φ_- (*phantom*), living on one spacetime. Both are governed by the same potential, the standard sextic form of Q-ball and solitonic boson-star studies [13–15],

$$V(|\Phi|^2) = \frac{1}{2}m^2|\Phi|^2 - \frac{1}{4}\lambda|\Phi|^4 + \frac{1}{6}\mu|\Phi|^6, \quad (1)$$

where m is the field mass and $\lambda, \mu > 0$ are self-coupling constants. Each term has a fixed job: the mass term makes the field oscillate, the attractive quartic ($-\lambda$) lets the field clump into localized, self-bound lumps, and the repulsive sextic ($+\mu$) stops the clumping short of collapse—with the quartic alone, which is critical in three dimensions, a lump either collapses or disperses. With all three terms the flat-space theory supports Q-ball-type solitons. (“Lump” is our generic word for such a self-bound blob of field; it becomes a “star” once gravity dresses it.) Self-binding is what lets the phantom star exist at all: a gravity-bound body cannot survive having its gravity reversed, but a field-bound lump merely gets dressed—compressed by attractive gravity in one sector, dilated by repulsive gravity in the other (Sec. III). Sharing one potential is deliberate, not economy: it makes the matter physics of the two sectors *identical*, so that the single sign introduced next is the only difference between them—and therefore the only possible cause of any asymmetry the simulations show.

The entire sign structure lives in one place: the Einstein equations are sourced by the *signed* sum of two canonical stress tensors,

$$G_{\mu\nu} = 8\pi (T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]), \quad (2)$$

while the field equations of *both* sectors keep their canonical form,

$$\square \Phi_{\pm} = \frac{\partial V}{\partial |\Phi|^2} \Phi_{\pm}. \quad (3)$$

In the 3+1 language the matter sources seen by normal observers—the energy density ρ and momentum density S_i that feed the ADM constraint and evolution equations—are accumulated per sector with a sign $\epsilon_{\pm} = \pm 1$:

$$\rho = \sum_{s=\pm} \epsilon_s \rho[\Phi_s], \quad S_i = \sum_{s=\pm} \epsilon_s S_i[\Phi_s], \quad (4)$$

and likewise for S_{ij} , each bracket being the standard scalar-field expression.

This structure delivers exactly the mass signature Bondi’s argument needs. How the phantom *moves* is set by its field equation (3), which is completely ordinary: the phantom field resists acceleration and falls toward gravitational wells like any other matter—in Bondi’s vocabulary, its *inertial* and *passive* masses are positive. How the phantom *gravitates* is set by the minus sign in Eq. (2): the field it sources is that of a negative mass, so its *active* mass is negative and it repels everything. In particular its Arnowitt–Deser–Misner (ADM) mass—the total mass a distant observer would infer from the far gravitational field—comes out negative.

Now place one star of each kind side by side. The canonical star attracts the phantom, which falls toward it; the phantom repels the canonical star, which retreats. Both accelerations point the same way, from phantom toward canonical, and the pair runs off together. This is the precise sense in which “Newton’s third law fails” here: the two forces point in the *same* direction rather than opposing each other. No conservation law breaks. The conserved momentum belongs to the total signed stress-energy of Eq. (4), in which the phantom’s momentum enters with a minus sign; the two growing momenta cancel, and the accelerating pair carries near-zero total momentum—the cancellation Forward exhibited for the Newtonian problem [5], here general-relativistic.

Two honest remarks. The phantom sector violates the null energy condition by construction; this is a model of *if such matter existed, what would gravity do*, in the spirit of phantom cosmology [8], not a claim that it exists. And our sign structure differs from Forward’s all-four-masses-negative matter [5]: here only the source sign flips. The two-body phenomenology is the same, because passive and inertial mass cancel—each body’s acceleration is set by the *other* body’s active mass, the property under test.

The model is implemented in `GRTeclyn` [20, 21, 23], evolving CCZ4 [19] plus the two complex fields, with Eq. (4) accumulated star by star in the initial-data solver `GRtresna` [22] so that one constraint solve can carry mixed-sign matter.

III. INITIAL DATA: GRAVITATIONALLY DRESSED SOLITONS

The task of this section is to turn the model into two stars on one initial slice: fix the couplings, solve each star as a spherical equilibrium, transfer the solved profiles onto the evolution grid, and solve the constraints for the pair. None of these steps is optional—the potential alone, without the gravitational dressing and the exact seeding below, produces no star that survives evolution. The failed campaigns that established this the hard way close the section.

A. Couplings

The potential (1) has three parameters: the mass $m = 1$ (our unit) and the self-couplings λ, μ . One more number needs a name before they can be chosen. A soliton of this theory oscillates coherently, $\Phi \propto e^{-i\omega t}$, and the frequency ω labels the solution family: $\omega = m$ is the free-field limit, the deficit $1 - \omega$ is the binding energy per unit mass, and lower ω means a more tightly bound, more massive lump.

Two combinations of the couplings control the physics. The dimensionless ratio λ^2/μ sets the *shape* of the potential and, through it, the lowest frequency a flat-space soliton can have: the classic Q-ball existence bound $\omega_{\min}^2 = \min_{|\Phi|} [2V/|\Phi|^2]$ [14] evaluates, for the potential (1), to $\omega_{\min} = \sqrt{1 - 3\lambda^2/16\mu}$, attained at the field value $\sqrt{3\lambda/4\mu}$ —which is also the interior (“thin-wall”) amplitude of a large soliton. All runs use $\lambda^2/\mu = 4.8$, so $\omega_{\min} = 0.316$ and the working frequency $\omega = 0.55$ sits comfortably inside the allowed range, at 45% binding. With the shape fixed, the remaining freedom—the overall strength of the couplings—sets that amplitude, and with it how strongly the lump gravitates: scaling λ and μ up together at fixed λ^2/μ makes the field fainter and its self-gravity weaker. The values used everywhere, $\lambda = 10240$ and $\mu \simeq 2.18 \times 10^7$ (amplitude $\simeq 0.019$), were forced on us: every more strongly gravitating choice produced stars that collapsed or dispersed (Sec. IIID), and these couplings are the first at which a dressed $\omega = 0.55$ star exists at all.

B. The dressed-star problem

The problem, stated plainly: a flat-space soliton solves the field equation with gravity switched off, and such an object cannot simply be placed on an initial slice. It fails as initial data, because the Hamiltonian and momentum constraints know about the field’s energy and are violated by a bare Q-ball; and it fails as an equilibrium, because once its own gravity acts the profile is no longer an eigenstate—seeded anyway, the star breathes, sheds, and dies (Sec. IIID documents exactly this). The star we need is solved with its gravity *on*: field and metric

adjusted together until the profile is the eigenstate of the very spacetime it curves. That is what “dressed” means here.

Dressing the soliton in its own gravity is the boson-star problem [16, 17] with a sextic self-interaction (see Ref. [18] for a review), posed for both source signs. Inserting the harmonic ansatz $\Phi = \varphi_0(r) e^{-i\omega t}$ and a conformally flat, maximally sliced metric $ds^2 = -\alpha^2 dt^2 + \psi^4 \delta_{ij} dx^i dx^j$ into Eqs. (2)–(3) reduces the problem to three coupled ordinary differential equations in r (the standard boson-star reduction [18]),

$$\begin{aligned} \psi'' + \frac{2}{r}\psi' &= -2\pi\psi^5\rho, \\ \alpha'' + \left(\frac{2}{r} + \frac{2\psi'}{\psi}\right)\alpha' &= 4\pi\psi^4\alpha(\rho + S), \\ \varphi_0'' + \left(\frac{2}{r} + \frac{\alpha'}{\alpha} + \frac{2\psi'}{\psi}\right)\varphi_0' &= \psi^4\left(\frac{dV}{d\varphi_0} - \frac{\omega^2}{\alpha^2}\varphi_0\right), \end{aligned} \quad (5)$$

with $\rho = \frac{1}{2}\omega^2\varphi_0^2/\alpha^2 + \frac{1}{2}\psi^{-4}\varphi_0'^2 + V$ and $\rho + S = 2\omega^2\varphi_0^2/\alpha^2 - 2V$. **The phantom star is obtained by flipping the sign of ρ and $\rho + S$ in the metric equations only;** the Klein–Gordon line keeps its canonical form, mirroring Eqs. (2)–(3). The result is a self-interaction-bound soliton with repulsive gravity: $\psi < 1$, $\alpha > 1$ in the core (reversed for the canonical star) and negative ADM mass.

We integrate Eqs. (5) by shooting; the solver and the $M(\omega)$ family scan behind Fig. 2 ship in the results pack. Solving at *fixed frequency* matters: heavy-branch sextic stars sit a fraction of a percent below the effective-potential top, so at fixed central amplitude the eigenvalue in ω is an exponentially thin needle, and a shooter on ω lands on the light branch (a request for the $\omega = 0.55$ star returned a $5\times$ lighter $\omega = 0.75$ star). Instead we bisect the central amplitude φ_c at fixed integration-frame frequency and outer-iterate until the rescaled frequency ω/α_∞ matches the request, to $\sim 10^{-5}$.

Figure 2 shows both families. The reference pair uses the two $\omega = 0.550$ stars: canonical ($\varphi_c = 0.019695$, $M_{\text{ADM}} = +0.063951$, central lapse 0.9773) and phantom ($\varphi_c = 0.019589$, $M_{\text{ADM}} = -0.076962$, central lapse 1.0249). Because both are taken at the *same* frequency, and the phantom branch is the heavier there, the reference pair is intrinsically unequal: $|M_-|/M_+ = 1.203$. For the equal- $|M|$ run of Sec. IV the phantom is instead solved at $\omega = 0.56598$, where $M_{\text{ADM}} = -0.063950$ matches the canonical star’s mass to one part in 10^5 .

C. From radial profiles to three-dimensional initial data

The solved profiles $\varphi_0(r)$, $\alpha(r)$, $\psi(r)$ are next transferred onto the Cartesian evolution grid, one star at each release position. Two details of this step mattered enormously (Sec. IIID). First, the seeded amplitude must be the star’s own solved φ_c —any rescaling de-tunes the eigenstate. Second, a boson star is stationary but not static: its phase rotates, so the field carries

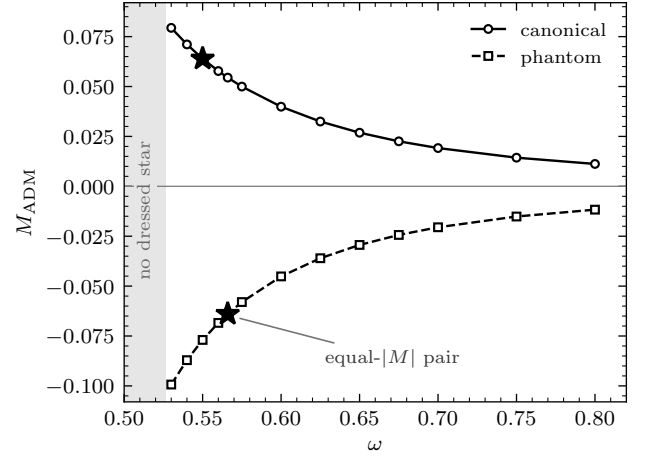


FIG. 2. ADM masses of the spherical stars against the field oscillation frequency ω , from Eqs. (5) at the working couplings: canonical stars ($M_{\text{ADM}} > 0$, circles) and phantom stars ($M_{\text{ADM}} < 0$, squares). Lower frequency means deeper binding and larger $|M_{\text{ADM}}|$; in the shaded region the solver finds no gravitationally dressed star. At equal frequency the phantom star is the heavier in magnitude (20% at $\omega = 0.55$, 4% at $\omega = 0.80$), its repulsive self-gravity spreading it over more volume. Stars ($\star$): the equal- $|M|$ pair of run PM-eq, matched to one part in 10^5 .

a $U(1)$ momentum, and the momentum conjugate to Φ is defined per unit *proper* time of the normal observers, $\Pi = -\alpha^{-1}\partial_t\Phi$ for zero shift. The harmonic phase therefore seeds

$$\Pi_{\text{im}} = -\frac{\omega}{\alpha(r)}\varphi_0(r); \quad (6)$$

a flat-space seed ($\alpha \equiv 1$) would start the star slightly off its eigenstate everywhere the lapse differs from one, which is why the star tables carry $\alpha(r)$. Each star is seeded from its own table with its own frequency; giving the $\omega = 0.56598$ star a global $\omega = 0.550$ phase velocity would likewise de-tune it.

GRTresna then solves the constraints for the pair (CT-TKHybrid scheme, sources summed with their signs star by star) to 0.1%; at an earlier 1% tolerance the leftover residual radiated away as a ring in χ that shook the stars’ envelopes. The evolution code keeps the solved metric and reconstructs the matter fields analytically from the same tables.

D. How the matter was made to hold together

None of the choices above were right on the first try. Before any runaway could be claimed, a single lump had to hold its own size; four campaigns failed that gate. Heavy lumps collapsed (min $\chi \rightarrow 0.009$ by $t \approx 25$), light lumps evaporated (rms $5 \rightarrow 30$ by $t \approx 50$), and lumps of intermediate weight did both—indicating the *shape* of the seed, not its weight. Four root causes, in order:

1. **The amplitude clamp.** The seeding script clamped the amplitude to the flat-space thin-wall value $\sqrt{3\lambda/4\mu}$, 4.5% below the solved eigenstate: an exact Q-ball shape at the wrong height is not a Q-ball, and it breathed, shed, and evaporated. *Fix:* seed at the solved φ_c .
2. **No dressed equilibrium existed.** The corrected flat-space seed still dispersed: a flat-space eigenstate is an equilibrium only in flat space, and the dressed family showed the target star did not exist at the original couplings. *Fix:* weaken the couplings to the working values and seed the dressed profile with its own lapse, Eq. (6).
3. **Two seeding paths disagreed silently.** The constraint solve used the tabulated star while the evolution’s matter reconstruction fell through to a Gaussian: the solved metric backed one object, the evolution started another. *Fix:* both paths resolve the same star, enforced to 10^{-10} by a regression test.
4. **The phantom path had never run.** An old veto downgraded exotic lumps to canonical matter whenever self-gravity was on—correct for a gravity-bound star, irrelevant for one bound by self-interaction. *Fix:* lift the veto. The first phantom star evolved matched its predicted profile, with the predicted mirror-image metric signature (χ just above unity).

Two residual effects were attributed rather than fixed: the $\pm 8\%$ radial breathing is resolution-intrinsic (resolving with $30\times$ tighter momentum residuals reproduced it crest for crest), and the rms-radius stream tracks the shed radiation bath (Sec. VI), not star health.

The exportable lesson: *verify the seed before believing the evolution.* Every run was checked at $t = 0$ against the independently integrated star solution; two of the four root causes were caught by that check, not by watching movies.

IV. EVOLUTION SETUP AND THE RUN MATRIX

With both stars in hand, the experiment is to release them at rest and watch the barycentres. Evolutions use **GRTEclyn** (one GPU per run, single rank): **CCZ4** [19] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, $1 + \log$ slicing and a Gamma-driver shift with $\eta = 1$, fourth-order spatial stencils, RK4 time stepping, Kreiss–Oliger dissipation $\sigma = 2$, and $\Delta t = 0.02 \Delta x$. The domain is $L = 64$ at base resolution $N = 128$ ($\Delta x = 0.5$), with Sommerfeld (radiative) outer boundaries and wave-extraction shells at $R = 8$ and 16. One mesh-refinement level is allowed (`max_level` = 1, χ -gradient tagging with threshold 0.02, regrid every 16 steps), but it first triggers at $t \approx 48$, after the measurement window, so every quantitative result was produced

TABLE I. The falsifiable run matrix. The sign structure of Sec. II predicts co-drift in $+x$ for the mixed pair, reversal under mirroring, a weaker runaway at wider separation, and *no* net drift for same-sector pairs and lone stars. Every run was solved to Ham $\leq 0.09\%$, Mom $\leq 0.08\%$ (Appendix A) and verified at $t = 0$ against the independently solved star profiles (within 0.2%). Drifts in code units at t_{end} ($t = 40$ for singles, 60 for pairs).

run	drift Φ_+	drift Φ_-	verdict
PM (+, −)	+3.35	+9.56	runaway
PM-eq (equal $ M $)	+2.94	+9.59	runaway
MP (mirror)	−3.23	−9.07	reversed
PP (+, +)	+0.16	—	null
MM (−, −)	—	+0.04	null
P (single +)	+0.06	—	null
M (single −)	—	+0.07	null
PM, $d_0=12$	+0.72	+2.78	weaker
PM, $d_0=16$	(biased)	+0.74	weaker

on the uniform grid (Appendix A). Stars are seeded at $x = \pm 4$ about the domain centre ($d_0 = 8$, about $1.6\times$ their rms radius of ~ 5), *at rest*—no boosts, no velocities, no driving of any kind—so any drift is gravitational.

Runs are named by content: PM is the mixed canonical–phantom pair, PP and MM the same-sector pairs, P and M the single stars, MP the mirrored pair (sectors swapped in place), and PM-eq the mixed pair rerun with the phantom’s mass magnitude matched to the canonical star’s, $|M_-| = M_+$ to one part in 10^5 (Fig. 2).

The primary diagnostic is the per-sector barycentre stream: field-weighted integrals, centroids, and rms radii for Φ_+ and Φ_- , sampled continuously. Constraint norms (Appendix A), confinement (peak amplitude, confined fraction, $\min \chi$), Ψ_4 modes, and energy-condition monitors are recorded alongside.

Table I is the experiment: each row is a falsifiable prediction of Sec. II’s sign structure—a same-sector pair or lone star that drifted like the mixed pair would kill the claim. PM-eq changes exactly one physical quantity (the phantom’s mass, by 17%); MP reflects the geometry; the wider separations probe the force law.

V. RESULTS

A. The runaway

Figures 3 and 4 carry the headline. In PM—canonical star at $x = +4$, phantom at $x = -4$, both at rest—*both* barycentres move in $+x$: by $t = 60$ the canonical star has moved +3.35 and the phantom +9.56, closing the gap from 8.00 to 1.79. The four controls, run under identical numerics, do not move (+0.16 and +0.04 for the same-sector pairs, +0.06/+0.07 for the singles), setting the noise floor; the mixed pair’s drift exceeds it by 50 to

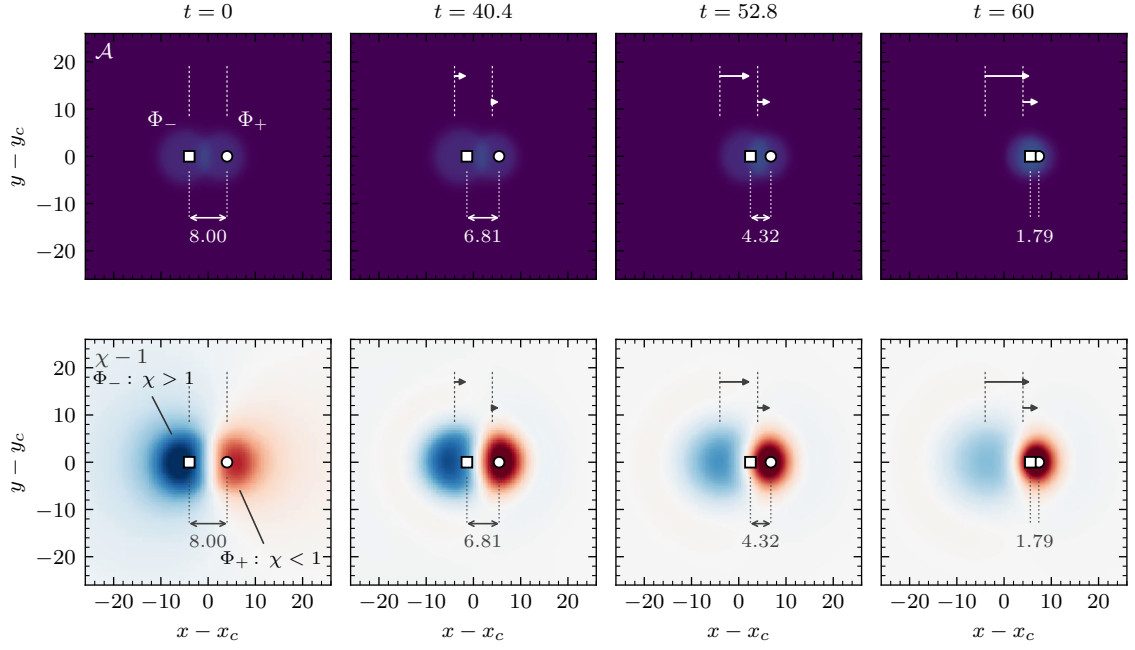


FIG. 3. The runaway, run PM, in the $z = 0$ plane. Top: scalar-field activity $\mathcal{A} = \sum_k (\varphi_k^2 + \Pi_k^2)^{1/2}$, both sectors. Bottom: $\chi - 1$; the phantom star is a hill ($\chi > 1$, blue), the canonical star a well ($\chi < 1$, red). Open markers: measured barycentres (circle canonical, square phantom); dashed verticals mark the release positions $x = \pm 4$, arrows the displacements; brackets give the separation, $8.00 \rightarrow 1.79$. Panels are individually color-scaled; the canonical well deepens from $\chi_{\min} = 0.98$ to 0.44 .

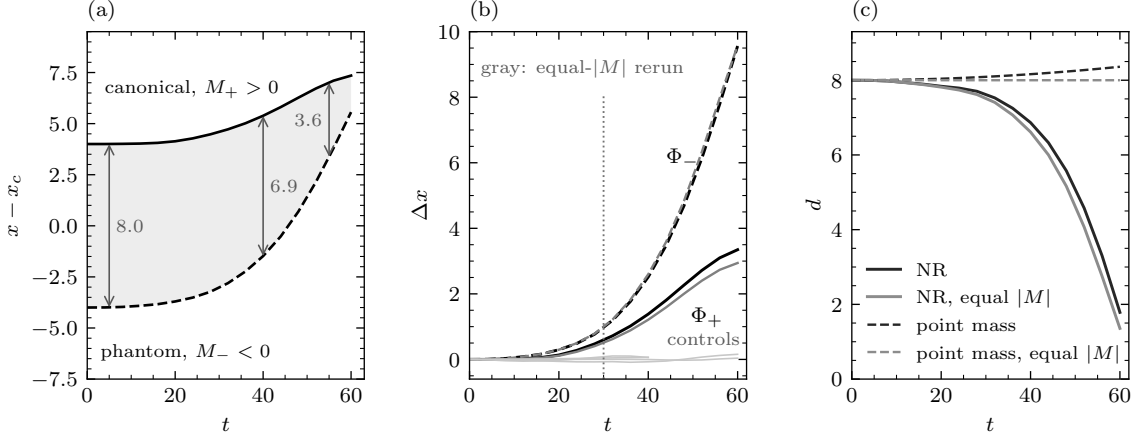


FIG. 4. (a) Barycentre worldlines of run PM: released at rest at $x = \pm 4$, both bodies move in $+x$, the phantom (dashed) chasing the canonical star (solid); the shaded band is the closing gap. (b) Drifts Δx , solid Φ_+ and dashed Φ_- : black, reference run; gray, PM-eq (Sec. VC); light gray, all four null controls. The dotted line marks the clean-window edge $t = 30$ (Sec. VI). (c) Separation against the matched point-mass integrations: the NR gap closes to near-contact where the model predicts slow growth.

150 times, with constraint norms at the level of the non-drifting controls throughout the window (Appendix A).

The barycentre *speeds* (Fig. 5) say the same thing kinematically: both bodies accelerate together from rest—a sustained pull, not an impulse. (Speeds, like everything else, are in code units with $c = 1$, so $v_x = 0.44$ means $0.44c$.) The canonical star reaches $v_x \approx 0.06$ at the window edge $t = 30$ and peaks near 0.12 ; the phantom passes 0.11 at $t = 30$ and reaches $v_x \approx 0.44$, a genuinely rela-

tivistic infall, as the pair nears contact. (The late decline of the canonical curve is the halo bias of Sec. VI, not a deceleration.)

Why does the phantom cover almost three times the distance? Not because of the mass imbalance—the run matrix contains the direct test. In PM-eq the mass magnitudes are matched to one part in 10^5 , and the phantom *still* covers 3.3 times the distance (+2.94 against +9.59; Table I). Theory agrees that the imbalance points the

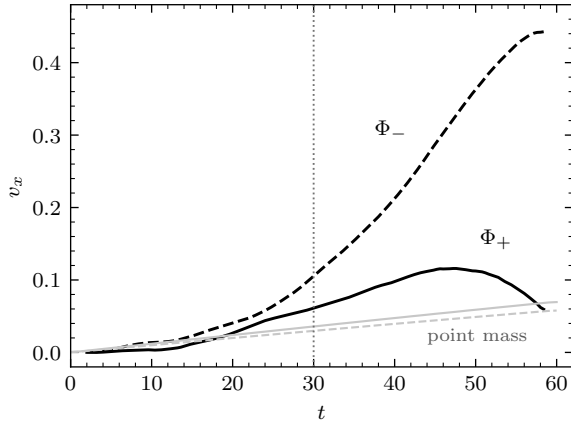


FIG. 5. Barycentre velocities in run PM, in code units ($c = 1$): centred differences of the continuously sampled barycentre stream over ± 2 time units (canonical solid, phantom dashed; light gray: the matched point-mass model). The dotted line marks the clean-window edge $t = 30$; beyond it the halo bias (Sec. VI) depresses the canonical curve.

TABLE II. The measured run against the Newtonian idealization. “NR” columns: run PM as measured. “model” columns: a point-mass Bondi integration—two point particles carrying the stars’ ADM masses (+0.0640 and -0.0770), released at rest at the same separation, each accelerated by the *other’s* active mass per Sec. II. Agreement at early times and the growing excess after $t \approx 30$ are discussed in Sec. VD; the clean window is $t \lesssim 30$ (Sec. VI).

t	drift Φ_+		drift Φ_-		separation	
	NR	model	NR	model	NR	model
10	+0.01	+0.06	+0.05	+0.05	7.96	8.01
20	+0.14	+0.24	+0.30	+0.20	7.84	8.04
30	+0.59	+0.54	+0.96	+0.45	7.63	8.09
40	+1.38	+0.96	+2.52	+0.79	6.87	8.16
60	+3.35	+2.13	+9.56	+1.77	1.79	8.36

wrong way: each body is accelerated by the *other’s* active mass, and $|M_-| > M_+$, so the point-mass model actually gives the *canonical* star the larger early drift (Table II). The asymmetry is an extended-body effect, quantified in Secs. VC and VD: the push on the canonical star stays Newtonian in the phantom’s mass to 0.6%, while the phantom’s fall runs up to $3\times$ ahead of the point-mass rate because its envelope sits deep inside the canonical star’s near field from $t = 0$ —a well that deepens further as the canonical star re-accretes its shed field ($\chi_{\min}$: $0.98 \rightarrow 0.44$)—whereas the phantom’s own shallow, self-diluting hill ($\chi \approx 1.001$) hands the canonical star no matching boost.

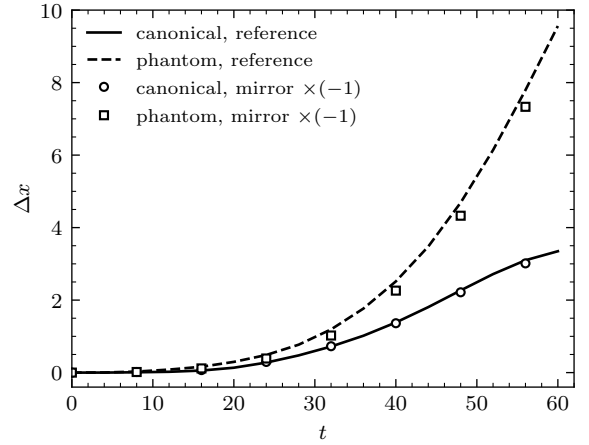


FIG. 6. Mirror control. Lines: reference run PM (canonical solid, phantom dashed); open markers: the mirrored run MP’s drift multiplied by -1 (circles canonical, squares phantom). Reflecting the configuration reflects the trajectory to a few percent.

B. The mirror control closes the artifact objection

A drift produced by grid anisotropy, boundary asymmetry, or diagnostic bias would not know which star is which; swapping the sectors must reverse the physics but not an artifact. It reverses the physics: flipped in sign, run MP reproduces the reference trajectory to 2.1%/15% (canonical/phantom) at $t = 30$ and 3.4%/5.2% at $t = 60$ (Fig. 6).

C. Gravity scales with the source

The strongest quantitative tests compare runs that differ in exactly one quantity.

The push scales with the phantom’s active mass. Between PM and PM-eq the phantom’s mass changes by the factor 0.831 and nothing else does; the canonical displacement responds in proportion (Table III): drift ratio 0.836 at $t = 20$, agreement to 0.6%. Barycentre biases largely cancel in this ratio, since both runs share the same separation and numerics.

The fall depends only on the other body. In the same pair of runs the canonical source is unchanged, and the phantom’s trajectory with it: drift ratio 1.002 at $t = 20$ and $t = 60$ (1.02–1.03 between). Two independent solves and evolutions, differing in the phantom’s own mass by 17%, reproduce its trajectory at the percent level—a physics statement and a reproducibility check at once.

D. Separation scaling: inverse-square once the stars separate

Table II shows the reference run departing from the point-mass model: the phantom overshoots increasingly

TABLE III. Does the push scale with its source? Between runs PM-eq and PM exactly one physical input changes: the phantom’s mass magnitude, smaller by the factor $0.06395/0.07696 = 0.831$. If the canonical star’s motion is gravity sourced by the phantom, its drift must shrink by that same factor. “NR ratio”: the measured canonical-star drift of PM-eq divided by that of PM; next to it, the same ratio predicted by the matched point-mass integrations, and the bare mass ratio itself.

t	NR ratio	point-mass ratio	mass ratio
20	0.836	0.832	0.831
30	0.859	0.834	0.831
40	0.876	0.837	0.831
60	0.879	0.843	0.831

TABLE IV. Ratio of the measured phantom drift to each run’s *own* matched point-mass integration. At $d_0 = 8$ the stars overlap from $t = 0$.

d_0	$t = 20$	$t = 30$	$t = 40$
8 (overlapping)	1.48	2.14	3.17
12	1.07	1.36	2.00
16	1.11	1.17	1.48

($1.5\times$ at $t = 20$, $3.2\times$ at $t = 40$) and the gap closes where the model grows it. PM-eq isolates the cause: with $|M|$ matched to 10^{-5} the model predicts identical drifts and a constant gap, yet the NR run gives $+2.94/+9.59$ and a gap closing to 1.35—not a mass-ratio effect. What remains is geometry: at $d_0 = 8$ these stars (rms radius ≈ 5) overlap from $t = 0$, the phantom’s envelope sitting inside the canonical near field where the pull exceeds M/r^2 .

The wider-separation runs measure this directly (Fig. 7). Fixed-time power-law fits are misleading here—the closing gap makes them report $n = 2.4\text{--}4.6$, measuring runaway feedback rather than the force law—so each run is compared against its *own* point-mass integration (Table IV). Where the point-mass idealization applies the measured drift is Newtonian to $\sim 10\%$ at early times; the $d_0 = 8$ excess is the extended-body overlap enhancement, falling away with separation.

E. The phantom sector is the stable one

Counter to intuition, the phantom star was the best-behaved object in the campaign (Table V): gravity is not the binding agent—the identical scalar self-interaction is—and the reversed sign is the *stabilizing* correction. Shed field around the canonical star is pulled back, deepening the well and feeding perturbations (χ $0.99 \rightarrow 0.85$ by $t = 40$); around the phantom it anti-gravitates and self-dilutes ($\chi = 1.0009 \pm 0.0002$ all run). The controls agree: MM keeps $\chi \approx 1.00$ while its material spreads; PP drives χ to 0.44 as it merges.

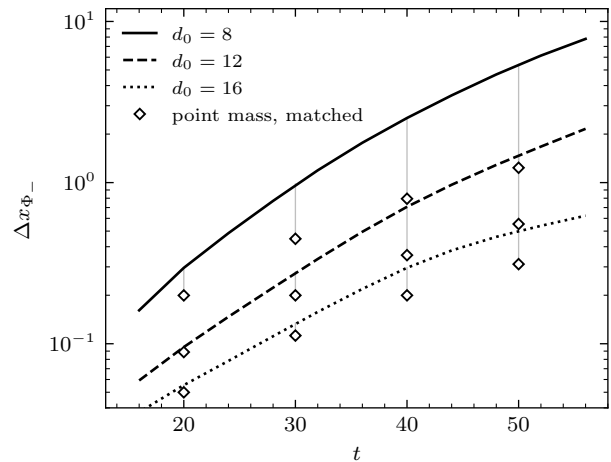


FIG. 7. Separation scaling of the phantom drift (the trustworthy sector; Sec. VI). Curves: NR runs at $d_0 = 8, 12, 16$; diamonds: each run’s *own* matched point-mass integration, tied to its curve at equal times (Table IV).

TABLE V. Single-star health, $t = 0 \rightarrow 40$.

metric	canonical	phantom
min χ	$0.989 \rightarrow 0.851$	$1.000 \rightarrow 1.001$
rms radius (radiation bath)	$5.05 \rightarrow 12.25$	$5.43 \rightarrow 6.25$
peak-amplitude breathing	$\pm 8\%$	$\pm 3\%$

VI. CAVEATS, AND THEIR MEASURED SCOPE

The barycentre includes the halo. It integrates core plus shed radiation; by $t = 60$ a third of the tracked canonical field has left the accounting region and the distribution has doubled in extent. The clean quantitative window is $t \lesssim 30$, where totals are stable to $\lesssim 7\%$ —exactly where the mass-scaling and reproducibility tests are strongest. Late-time drifts are upper bounds on core displacement; the qualitative result holds across the whole run.

The canonical centroid is biased low. At $d_0 = 16$ it drifts *backwards* (-0.34 by $t = 60$): the phantom’s repulsion pushes canonical matter out through the $+x$ boundary, and losing the leading edge drags the centroid back. Canonical drifts are therefore underestimates everywhere (including the reference $+3.35$), worst where the true displacement is small—wide gaps, early times. The phantom sector, which gains tracked weight rather than losing it, is the trustworthy one, and Sec. VD uses it; the mass-scaling ratio compares equal-separation runs, so the bias largely cancels there. A core-weighted centroid immune to the halo is built for the follow-up.

The radiation bath cannot leave. Massive-field radiation reflects from massless-wave (Sommerfeld) boundaries and washes back over the stars; this caps the mixed runs at $t = 60$. A sponge layer is the fix.

Resolution. Every run used a single resolution ($N = 128$), single rank, effectively unigrid (Appendix A). The only sensitivity probe—the $30\times$ tighter constraint solve, which changed nothing—addresses the solve, not the evolution; a proper convergence series is future work.

VII. DISCUSSION AND CONCLUSION

What is established. We posed Bondi’s positive-negative mass pair as a full initial-value problem, and it behaved exactly as he argued in 1957. The main results, restated:

- (1) *The runaway is real in full general relativity.* Released at rest, the mixed pair—and only the mixed pair—self-accelerates, the phantom chasing the canonical star: displacements $+3.35$ and $+9.56$ by $t = 60$, which is 50 to 150 times the residual motion of the single-star and same-sign controls, and which reverses under mirror reflection to a few percent.
- (2) *The effect responds to its source the way gravity must.* The push on the canonical star scales with the phantom’s active mass to 0.6%; the phantom’s fall is set by the canonical star alone, reproducing across independent solves to 0.2%; at wider separations the drift converges to the matched point-mass prediction to about 10%.
- (3) *Momentum is conserved throughout.* The phantom’s momentum enters the signed bookkeeping with a minus sign, so the accelerating pair carries near-zero total momentum: Forward’s accounting survives the transition from point particles to dynamical, radiating, finite-size matter.
- (4) *A stable star with negative ADM mass exists* as constraint-satisfying initial data and survives evolution—to our knowledge a first, and possibly the most reusable artifact of the campaign, since the phantom star proved the best-behaved object in it.

What is not established. A *constant-gap* runaway—the textbook picture of the pair accelerating forever at fixed separation—requires point-like separation. At $d_0 = 8$ the stars overlap and close to near-contact by $t = 60$; at $d_0 = 12$ – 16 the gap-closure is milder but the window ends before the asymptotic behaviour is visible. The clean test needs the wider separations *and* a longer window (a sponge layer), or more compact stars (higher ω , where the sector mass asymmetry also shrinks; Fig. 2). Every mixed run was stopped before contact; what a canonical-phantom collision produces—including whether a horizon forms—is untouched. And nothing here bears on whether NEC-violating matter exists; the model is a probe of the gravitational dynamics, not a proposal for a propulsion system [5].

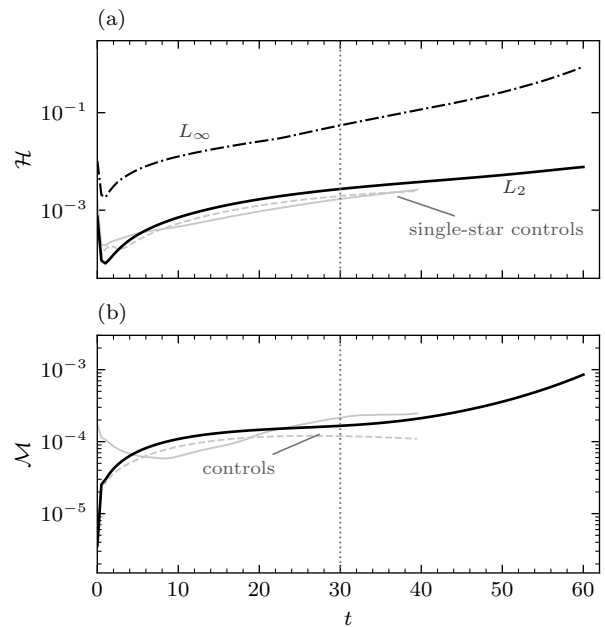


FIG. 8. Constraint monitoring: reference run PM (black) against the single-star controls (light gray; solid canonical, dashed phantom). (a) Hamiltonian constraint $\mathcal{H}$: composite L_2 norms (solid) and the reference run’s L_∞ (dash-dotted). (b) Momentum constraint $\mathcal{M}$, L_2 . The equal- $|M|$ rerun lies indistinguishably under the black curves and is omitted; the dotted line marks the clean-window edge $t = 30$.

Next steps. Core-weighted centroids and a per-sector momentum-balance diagnostic (the direct Bondi bookkeeping check) are built and waiting; the separation series should be re-run with them before the scaling numbers are quoted as final. A sponge layer unlocks $t \gg 60$ and the constant-gap test; the contact end state and a convergence series follow.

REPRODUCIBILITY

All time series, star tables, parameter files, code patches, analysis scripts, and the campaign journal behind every number and figure are packed, with machine paths scrubbed, in the repository’s results tree (`results/bondi-dipole-runaway/`); the figures regenerate from that pack by a single script. Run labels map to pack directories as `PM = pair_pm`, `PM-eq = pair_pm_eqm`, `MP = pair_mp_mirror`, `PP = pair_pp`, `MM = pair_mm`, `P = single_p`, `M = single_m`; the star ODE solver of Sec. III and the $M(\omega)$ scan behind Fig. 2 are `analysis/star_family_scan.py` and the `boson_star_ode` module it drives. Initial data: GRtresna [22]; evolution: GRteclyn [21], the AMReX-based [23] GPU port of GRChombo [20].

Appendix A: Constraint behaviour

Initial data. GRTresna’s nonlinear iteration contracts monotonically on every run; for PM the Hamiltonian/momentum error falls 100% $\rightarrow$ 0.083%/0.074% in seven iterations, and every production run exits at $\leq 0.09\%/ \leq 0.08\%$ —the gate quoted in Table I, in the solver’s own grid and normalization.

Evolution. The constraints are re-evaluated on the evolution grid every $\Delta t = 0.5$ and reduced to composite norms over the hierarchy (covered coarse cells dropped, outermost four cells excluded so boundary violation is not misread as interior error). Figure 8 shows the record, in absolute code units: over an overwhelmingly vacuum domain a global *relative* normalization is dominated by near-zero denominators, so the yardstick is the controls run under identical numerics.

Three features carry the message. The birth transient:

re-evaluating with fourth-order stencils after the analytic matter reconstruction (Sec. III) gives $L_2(\mathcal{H}) \simeq 7 \times 10^{-4}$, relaxing within $t \lesssim 1$ to the truncation floor 8×10^{-5} . Secular growth: $L_2(\mathcal{H})$ reaches 2.7×10^{-3} at the window edge $t = 30$ and 7.7×10^{-3} at $t = 60$, $L_\infty(\mathcal{H})$ 0.05 and 0.9, concentrated in the deepening canonical well as the pair nears contact and the trapped bath washes back (Sec. VI); $L_2(\mathcal{M})$ stays below 2×10^{-4} through $t = 40$. This growth is the quantitative face of the $t \lesssim 30$ window. And decisively: through that window the mixed run’s violation is within a factor 1.6 of the non-drifting single-star controls (2.7×10^{-3} against 1.7×10^{-3} at $t = 30$) while the drifts differ by 50 to 150 times—the drift correlates with the sign structure, not the constraint error.

The single refinement level (χ -gradient tagging, regrid every 16 steps) first activates at $t \approx 47.5$ in the mixed runs and never in the singles: every quantitative result was produced on the uniform $\Delta x = 0.5$ grid.

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